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A simplified multi-class support vector machine with reduced dual optimization

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ABSTRACT

Support vector machine (SVM) was initially designed for binary classification. To extend SVM to the multi-class scenario, a number of classification models were proposed such as the one by Crammer and Singer (2001). However, the number of variables in Crammer and Singer's dual problem is the product of the number of samples (l) by the number of classes (k), which produces a large computational complexity. This paper presents a simplified multi-class SVM (SimMSVM) that reduces the size of the resulting dual problem from $l \times k$ to l by introducing a relaxed classification error bound. The experimental results demonstrate that the proposed SimMSVM approach can greatly speed-up the training process, while maintaining a competitive classification accuracy.

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1. Introduction

Support vector machine (SVM) originally separates the binary classes ($k = 2$) with a maximized margin criterion (Cortes and Vapnik, 1995). However, real world problems often require the discrimination for more than two categories. Thus, the multi-class pattern recognition has a wide range of applications including Optical Character Recognition (Mori et al., 1995), Intrusion Detection (Khan et al., 2007), Speech Recognition (Ganapathiraju et al., 2004), and Bioinformatics (Baldi and Pollastri, 2002). In practice, the multi-class classification problems ($k > 2$) are commonly decomposed into a series of binary problems such that the standard SVM can be directly applied. Two representative ensemble schemes are one-versus-rest (1VR) (Vapnik, 1998) and one-versus-one (1V1) (Kreßel, 1999) approaches. Both one-versus-rest and one-versus-one are special cases of the Error Correcting Output Codes (ECOC) (Dietterich and Bakiri, 1995) which decomposes the multi-class problem into a pre-defined set of binary problems. A main issue of this approach is to construct a good ECOC matrix. Another proposals (Vapnik, 1998; Weston and Watkins, 1999; Bredensteiner and Bennett, 1999; Guermeur, 2002; Crammer and Singer, 2001) are to directly address the multi-class problem in one single optimization processing. This kind of models combines multiple binary-class optimization problems into one single objective function and simultaneously achieves classification of multiple classes. However, a larger computational complexity is required for the size of the resulting Quadratic Programming (QP) problem.

Moreover, Szedmak et al. (2004) proposed a multi-class model for L1-norm SVM. In their formulation, the one-versus-rest framework is used. A potential problem with one-versus-rest is that when the number of classes is large, each binary classification becomes highly unbalanced. The unbalanced classification problem occurs when there are many more samples of some classes than others. In this case, standard classifiers tend to be overwhelmed by the large-scale classes and ignore the small ones. The SVM algorithm is constructing a separating hyperplane with the maximal margin. Since only support vectors are used for classification and many majority samples far from the decision boundary can be removed, SVM can be more accurate on moderately unbalanced data. However, SVM is sensitive to high unbalanced classification since it is prone to generating a classifier that has a strong estimation bias towards the majority class and would give a bad accuracy in the classification performance for the minority class, which is discussed in the work of Chawla et al. (2004) and Tang et al. (2009). Wang and Shen (2007) proposed a method that circumvents the difficulties of one-versus-rest by treating multiple classes jointly. Suykens and Vandewalle (1999b) extended the LS-SVM methodology (Suykens and Vandewalle, 1999a) to the multi-class case. A drawback of LS-SVM is that its solution is constructed from most of the training examples, which is referred to as the non-sparseness problem. Xia and Li (2008) presented a new multi-class LS-SVM algorithm whose solution is sparse in the weight coefficient of support vectors. Fung and Mangasarian (2005) followed the PSVM (Proximal SVM) (Fung and Mangasarian, 2001) idea and applied it to the multi-class problem. This approach is closely aligned with the one-versus-rest method. For each decomposed subproblem, the solution is similar to its binary case (PSVM) which classifies new samples by assigning them to the

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closer of the two parallel planes that are pushed apart as far as possible.

This paper focuses on the multi-class classification and presents a simplified multi-class support vector machine (SimMSVM). SimMSVM gives a direct solution for training multi-class predictors. Following Crammer2001s work, this paper introduces a relaxed classification error bound. By doing so, the resulting dual problem only involves l variables, where l is the size of training samples. That is, solving a single l-variable QP is enough for a multi-class classification task. We then investigate the theoretical properties of our loss function, including the Fisher consistency issue. The experimental results validate that the proposed simplified model achieves significant speed-up compared with its original model (Crammer and Singer, 2001). At the same time, the proposed SimMSVM approach outperforms the other state-of-the-art multi-class SVM algorithms in terms of computational time while maintaining competitive classification accuracies.

The remainder of the paper is organized as follows. Section 2 describes the related work for the multi-class and reduced SVM. Section 3 gives the detail of the proposed SimMSVM method. The characters of the proposed SimMSVM are presented in Section 4. We empirically evaluate the proposed method in Section 5. Section 6 gives the conclusions.

2. Related work

In this section, we provide a description on several implemented methods including the one-versus-rest approach (Vapnik, 1998), the one-versus-one approach (Knerr et al., 1990), Weston and Watkins' multi-class SVM (Weston and Watkins, 1999), and Crammer and Singer's multi-class SVM (Crammer and Singer, 2001) for solving a multi-class classification problem. Furthermore, we give the review for the work reducing the computational complexity of kernel-based learning since the proposed method also aims to simplify the computation for multi-class SVM.

2.1. SVM classification

SVM seeks to find the optimal separating hyperplane between binary classes by following the maximized margin criterion (Cortes and Vapnik, 1995). Given training vectors $\boldsymbol{x}_i \in \mathcal{R}^d,\ i = 1,\ldots,l$, in two classes, and the label vector $\boldsymbol{y} \in \{1,-1\}^l$, the support vector technique requires the solution of the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{w}\in\mathcal{H},b\in\mathcal{R},\xi_i\in\mathcal{R}} \quad & \frac{1}{2}\boldsymbol{w}^T\boldsymbol{w} + C\sum_{i=1}^{l}\xi_i, \\ \text{subject to} \quad & y_i(\boldsymbol{w}^T\boldsymbol{\varphi}(\boldsymbol{x}_i) + b) \geqslant 1 - \xi_i, \\ & \xi_i \geqslant 0, \quad i = 1,\ldots,l, \end{aligned} \tag{1}$$

where $\boldsymbol{w} \in \mathcal{R}^d$ is the weight vector, $C \in \mathcal{R}_+$ is the regularization constant, and the mapping function $\boldsymbol{\varphi}$ projects the training data into a suitable feature space $\mathcal{H}$ so as to allow for nonlinear decision surfaces. A common method to solve (1) is through its dual:

$$\begin{aligned} \min_{\boldsymbol{\alpha}\in\mathcal{R}^l} \quad & \frac{1}{2}\boldsymbol{\alpha}^T(\boldsymbol{K}\odot(\boldsymbol{y}\boldsymbol{y}^T))\boldsymbol{\alpha} - \boldsymbol{e}^T\boldsymbol{\alpha}, \\ \text{subject to} \quad & \boldsymbol{0} \leqslant \boldsymbol{\alpha} \leqslant C\boldsymbol{e}, \quad \boldsymbol{y}^T\boldsymbol{\alpha} = 0, \end{aligned} \tag{2}$$

where $\boldsymbol{e} \in \mathcal{R}^l$ is a vector of all 1s, $\boldsymbol{K} \in \mathcal{R}^{l\times l}$ is the kernel matrix, and $\odot$ denotes the Hadamard-Schur (elementwise) product. Symbols in bold represent matrices or vectors, as particular case, the symbol $\boldsymbol{0}$ represents the vector with all components set to 0.

Instead of (1), Mangasarian and Musicant (1999) proposed a new SVM formulation:

$$\begin{aligned} \min_{\boldsymbol{w}\in\mathcal{H},b\in\mathcal{R},\xi_i\in\mathcal{R}} \quad & \frac{1}{2}\left(\boldsymbol{w}^T\boldsymbol{w} + b^2\right) + C\sum_{i=1}^{l}\xi_i, \\ \text{subject to} \quad & y_i(\boldsymbol{w}^T\boldsymbol{\varphi}(\boldsymbol{x}_i) + b) \geqslant 1 - \xi_i, \\ & \xi_i \geqslant 0, \quad i = 1,\ldots,l, \end{aligned} \tag{3}$$

whose dual becomes a bound-constrained problem:

$$\begin{aligned} \min_{\boldsymbol{\alpha}\in\mathcal{R}^l} \quad & \frac{1}{2}\boldsymbol{\alpha}^T\left((\boldsymbol{K} + \boldsymbol{e}\boldsymbol{e}^T)\odot(\boldsymbol{y}\boldsymbol{y}^T)\right)\boldsymbol{\alpha} - \boldsymbol{e}^T\boldsymbol{\alpha}, \\ \text{subject to} \quad & \boldsymbol{0} \leqslant \boldsymbol{\alpha} \leqslant C\boldsymbol{e}. \end{aligned} \tag{4}$$

The motivation behind (3) is that, it is supposed easier to handle its dual problem (4) without linear constraint. Hsu and Lin (2002b) demonstrate that (3) is an acceptable formulation in terms of generalization errors though an additional term $b^2/2$ is added to the objective. This method is implemented in the software BSVM (Hsu and Lin, 2008) which is proved to have fast convergence in the literature (Hsu and Lin, 2002b).

2.2. One-versus-rest approach

The one-versus-rest (1VR) approach (Vapnik, 1998) constructs k separate binary classifiers for k-class classification. The mth binary classifier is trained using the data from the mth class as positive examples and the remaining $k-1$ classes as negative examples. During test, the class label is determined by the binary classifier that gives maximum output value. A major problem of the one-versus-rest approach is the imbalanced training set. Suppose that all classes have an equal size of training examples, the ratio of positive to negative examples in each individual classifier is $\frac{1}{k-1}$. In this case, the symmetry of the original problem is lost.

2.3. One-versus-one approach

Another classical approach for multi-class classification is the one-versus-one (1V1) or pairwise decomposition (Knerr et al., 1990). It evaluates all possible pairwise classifiers and thus induces $k(k-1)/2$ individual binary classifiers. Applying each classifier to a test example would give one vote to the winning class. A test example is labeled to the class with the most votes. The size of classifiers created by the one-versus-one approach is much larger than that of the one-versus-rest approach. However, the size of QP in each classifier is smaller, which makes it possible to train fast. Moreover, compared with the one-versus-rest approach, the one-versus-one method is more symmetric. Platt et al. (2000) improved the one-versus-one approach and proposed a method called Directed Acyclic Graph SVM (DAGSVM) that forms a tree-like structure to facilitate the testing phase. As a result, it takes only $k-1$ individual evaluations to decide the label of a test example.

2.4. Weston and Watkins' multi-class SVM

Instead of creating several binary classifiers, a more natural way is to distinguish all classes in one single optimization processing, as proposed by Vapnik (1998), Weston and Watkins (1999), and Bredensteiner and Bennett (1999). For a k-class problem, these methods design a single objective function for training all k-binary SVMs simultaneously and maximize the margins from each class to the remaining ones. Here, we take the method by Weston and Watkins as an example. Given a labeled training set represented by $\{(\boldsymbol{x}_1,y_1),\ldots,(\boldsymbol{x}_l,y_l)\}$ of cardinality l, where $\boldsymbol{x}_i \in \mathcal{R}^d$ and $y_i \in \{1,\ldots,k\}$, the formulation proposed in (Weston and Watkins, 1999) is given as follow:

$$\begin{aligned} \min_{\boldsymbol{w}_m \in \mathcal{H}, \boldsymbol{b} \in \mathcal{R}^k, \xi \in \mathcal{R}^{l \times k}} \quad & \frac{1}{2} \sum_{m=1}^{k} \boldsymbol{w}_m^T \boldsymbol{w}_m + C \sum_{i=1}^{l} \sum_{t \neq y_i} \xi_{i,t}, \\ \text{subject to} \quad & \boldsymbol{w}_{y_i}^T \boldsymbol{\varphi}(\boldsymbol{x}_i) + b_{y_i} \geqslant \boldsymbol{w}_t^T \boldsymbol{\varphi}(\boldsymbol{x}_i) + b_t + 2 - \xi_{i,t}, \\ & \xi_{i,t} \geqslant 0, \\ & i = 1, \ldots, l, \quad t \in \{1, \ldots, k\} \setminus y_i. \end{aligned} \tag{5}$$

The resulting decision function is

$$\operatorname{argmax}_m f_m(\boldsymbol{x}) = \operatorname{argmax}_m (\boldsymbol{w}_m^T \boldsymbol{\varphi}(\boldsymbol{x}) + b_m). \tag{6}$$

The main disadvantage of this approach is that the computational time may be very high due to the enormous size of the resulting QP.

2.5. Crammer and Singer's multi-class SVM

Crammer and Singer (2001) presented another "all-together" approach by solving the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{w}_m \in \mathcal{H}, \xi \in \mathcal{R}^l} \quad & \frac{1}{2} \sum_{m=1}^{k} \boldsymbol{w}_m^T \boldsymbol{w}_m + C \sum_{i=1}^{l} \xi_i, \\ \text{subject to} \quad & \boldsymbol{w}_{y_i}^T \boldsymbol{\varphi}(\boldsymbol{x}_i) - \boldsymbol{w}_t^T \boldsymbol{\varphi}(\boldsymbol{x}_i) \geqslant 1 - \delta_{y_i,t} - \xi_i, \\ & i = 1, \ldots, l, \quad t \in \{1, \ldots, k\}, \end{aligned} \tag{7}$$

where $\delta_{i,j}$ is the Kronecker delta (defined as 1 for $i = j$ and as 0 otherwise). The resulting decision function is

$$\operatorname{argmax}_m f_m(\boldsymbol{x}) = \operatorname{argmax}_m \boldsymbol{w}_m^T \boldsymbol{\varphi}(\boldsymbol{x}). \tag{8}$$

Note that the constraints $\xi_i \geqslant 0$, $i = 1, \ldots, l$, are implicitly indicated in the margin constraints of (7) when t equals y_i. In addition, (7) focuses on classification rule (8) without the bias terms b_m, $m = 1, \ldots, k$. A non-zero b_m can be easily modeled by adding an additional constant feature to each $\boldsymbol{x}$ (see, e.g. Mangasarian and Musicant, 2001).

2.6. Reducing the complexity of kernel-based methods

It is well-known that Williams and Seeger (2001) developed a Nyström technique that approximates a $l \times l$ kernel matrix K by sampling only p ($p \ll l$) rows and corresponding columns from K and thus efficiently decreases the computational complexity from $o(l^3)$ to $o(p^2 l)$. In practice, to achieve an exact kernel matrix approximation, a large number of rows (columns) for p should be selected. But more selected rows (columns) would result in a high computation complexity. Therefore, a key problem for the Nyström technique is how to define a better sampling scheme for the selected columns so as to obtain a good balance between the effectiveness and efficiency. First, Williams and Seeger (2001) adopt the uniform sampling without replacement so as to perform the column selection. Then Mohri and Talwalkar (2009) proved the uniform sampling technique to be much efficient in time and space complexity. Further more sophisticated non-uniform sampling techniques were developed. Drineas and Mahoney (2005) utilized the spectral and Frobenius norm to develop a non-uniform sampling scheme, which was based on the data-dependent non-uniform probability distribution. Zhang et al. (2008) presented a Nyström-related error analysis and gave that the error bound could result in a k-mean cluster algorithm for the sampled columns. It is known that the more columns are sampled, the more accurate approximation is got in the Nyström technique (Williams and Seeger, 2001). Thus, Kumar et al. (2009) gave an ensemble Nyström method which sampled more columns in a special way so as to improve performance. The ensemble technique consisted of the n_e Nyström approximation with p columns sampled in every approximation. Then the n_e approximation results were integrated. This method sampled $n_e \times p$ columns and thus effectively improved the approximating accuracy. Recently Li et al. (2010) introduced a randomized low-rand approximation method to perform the Singular Value Decomposition (SVD) for the rank-k sub-matrix instead of the $p \times p$ matrix W. They then combined the Nyström method with the randomized SVD so as to further decrease the computational complexity in kernel-based methods.

3. The proposed simplified multi-class SVM (SimMSVM)

3.1. Rationale of SimMSVM

Although Crammer and Singer's multi-class SVM gives a compact set of constraints, the number of variables in its dual problem is still $l \times k$ (Crammer and Singer, 2001). This value may explode even for small datasets. For instance, an English letter recognition problem with 2600 samples (100 samples per letter) would require solving a QP of size 2600×26, which will result in a large computational complexity. In this section, we follow Crammer and Singer's work and propose a simplified method for relaxing its constraints. By doing so, solving one single l-variable QP is enough for a multi-class classification task.

Before we describe our algorithms, we first compare the loss function of the above two "all-together" approaches. For a training example $\mathbf{x}_i$, we let

$$\zeta_{i,m} = 1 - f_{y_i}(\boldsymbol{x}_i) + f_m(\boldsymbol{x}_i) \tag{9}$$

for $m \in \{1, \ldots, k\} \setminus y_i$. If $\zeta_{i,m} > 0$, it depicts the pairwise margin violation between the true class y_i and some other class m. In Weston and Watkins' work, their loss function adds up all positive margin violation ($\zeta_{i,m} > 0$),

$$\xi_i^{(1)} = \sum_{m \neq y_i} [\zeta_{i,m}]_+, \tag{10}$$

where $[\cdot]_+ \equiv \max(\cdot, 0)$. In the original work proposed by Weston and Watkins (1999), the term ζ adopts the "2" rather than "1" as follows

$$\zeta_{i,m} = 2 - f_{y_i}(\boldsymbol{x}_i) + f_m(\boldsymbol{x}_i).$$

Here in order to compare the work proposed by Weston and Watkins (1999) with the other methods consistently, we scale the "2" into "1", i.e. adopt the Eq. (9). As for Crammer and Singer's approach, sample loss counts only the maximum positive margin violation

$$\xi_i^{(2)} = \left[\max_{m \neq y_i} \zeta_{i,m} \right]_+. \tag{11}$$

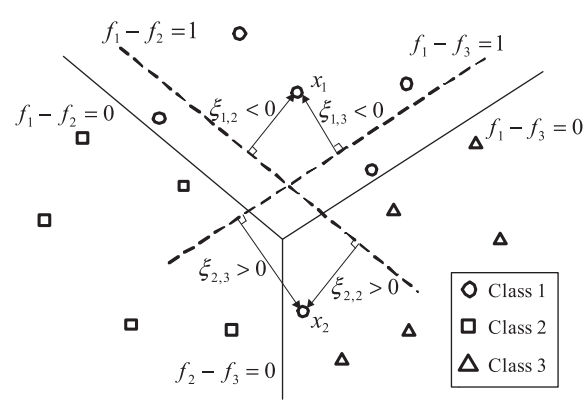

Fig. 1. Multi-class classification visualization. Class 1, 2 and 3 are represented by circles, rectangles, and triangles, respectively. The bold lines represent some possible class boundaries. The two dash lines are two positive margins for the first class. The pairwise margin violations for two examples from the first class, $\zeta_{1,2}$ and $\zeta_{1,3}$ for $\boldsymbol{x}_1$, and $\zeta_{2,2}$ and $\zeta_{2,3}$ for $\boldsymbol{x}_2$, are depicted in the figure.

Fig. 1 gives a multi-class classification graphical illustration, where three classes are represented as circles, rectangles, and triangles, respectively. The bold lines represent some possible decision boundaries. We plot the two positive pairwise margins for the first class (shown in dash lines). For a correctly classified example $\boldsymbol{x}_1$, we have $\xi_1^{(1)} = 0$ and $\xi_1^{(2)} = 0$, i.e. no loss counted, since both $\zeta_{1,2}$ and $\zeta_{1,3}$ are negative. On the other hand, for an examplebi x_2 that violates two margin bounds ($\zeta_{2,2}$, $\zeta_{2,3} > 0$), both methods generate a loss.

In order to reduce the problem size, the number of constraints should be proportional to l instead of $l \times k$. To construct our multi-class predictors, we introduce the following relaxed bound

$$\xi_i^{(3)} = \left[\frac{1}{k-1}\sum_{m=1, m\neq y_i}^{k} \zeta_{i,m}\right]_+. \tag{12}$$

That is, we suffer a loss which is linearly proportional to the average of directed distances between an example and its pairwise margins. For the above example shown in Fig. 1, the proposed loss (12) yields no loss for $\boldsymbol{x}_1$ as the other two loss functions.

To clearly analyze the proposed method, we compare the three loss functions (10)–(12). The loss (10) adds up all positive margin violation ($\zeta_{i,m} > 0$). The loss (11) counts only the maximum positive margin violation. The loss (12) represents a linear proportion to the average of directed distances between a sample and its pairwise margins. Thus, the relationship between these losses are

$$\xi_i^{(1)} \geqslant \xi_i^{(2)} \geqslant \xi_i^{(3)}. \tag{13}$$

Fig. 1 gives a visual example for the three loss functions (10)–(12). According to their definition, the corresponding losses for the sample x_2 that violates two margin bounds in Fig. 1 are

$$\begin{aligned}
\xi_2^{(1)} &= \zeta_{2,2} + \zeta_{2,3} \\
\xi_2^{(2)} &= \max\{\zeta_{2,2}, \zeta_{2,3}\} = \zeta_{2,3} \\
\xi_2^{(3)} &= (\zeta_{2,2} + \zeta_{2,3})/2
\end{aligned}$$

From the inequality (13), the relationship in Fig. 1 for the losses are

$$\xi_2^{(1)} \geqslant \xi_2^{(2)} \geqslant \xi_2^{(3)}.$$

In order to clearly explore the chosen loss function, we rewrite our relaxed loss function (12) as

$$\xi_i = \left[1 - f_{y_i}(\boldsymbol{x}_i) + \frac{1}{k-1}\sum_{m\neq y} f_m(\boldsymbol{x}_i)\right]_+. \tag{14}$$

In other words, the proposed method will yield no loss for a training example $\boldsymbol{x}_i$ if $f_{y_i}(\boldsymbol{x}_i) \geqslant 1 + \frac{1}{k-1}\sum_{m\neq y_i} f_m(\boldsymbol{x}_i)$. Although we cannot guarantee if $\xi_i = 0$ that $f_{y_i}(\boldsymbol{x}_i) > f_m(\boldsymbol{x}_i)$ for all $m \neq y_i$, it is most likely that f_{y_i} will be the maximum value of $f_1, \ldots, f_k$. To further explore this statement, we empirically study the problem in all of the ten datasets. Table 1 gives the rates that f_{y_i} be the maximum value of $f_1, \ldots, f_k$ and thus validate our claim about the proposed loss function. In this sense, the (14) is reasonable as a classification criterion.

In theory, our relaxed loss function (14) may incur more classification error since it fails to upper-bound the 0–1 loss. In practice, we admit some tolerable classification error in order to obtain a significant speed-up in the training process. Our experimental result in Section 5 demonstrate that, SimMSVM gives competitive accuracies with the other multi-class SVM approaches on real world datasets. On the other hand, the multi-class hinge losses of Weston and Watkins (1999) and Crammer and Singer (2001) both satisfy as the upper bound of 0–1 loss. However, in practice, they all involve a QP of size $l \times k$ and thus cause a high computational complexity for a relatively large number of classes.

3.2. Architecture of SimMSVM

Induced by the above error bound, our unbiased primal problem comes up as follows

$$\begin{aligned}
\min_{\boldsymbol{w}_m \in \mathcal{H}, \xi \in \mathcal{R}^l} \quad & \frac{1}{2}\sum_{m=1}^{k} \boldsymbol{w}_m^T \boldsymbol{w}_m + C\sum_{i=1}^{l} \xi_i, \\
\text{subject to} \quad & \boldsymbol{w}_{y_i}^T \boldsymbol{\varphi}(\boldsymbol{x}_i) - \frac{1}{k-1}\sum_{m\neq y_i} \boldsymbol{w}_m^T \boldsymbol{\varphi}(\boldsymbol{x}_i) \geqslant 1 - \xi_i, \\
& \xi_i \geqslant 0, \quad i = 1, \ldots, l.
\end{aligned} \tag{15}$$

Note that the number of margin constraints (15) has already been reduced to l. To solve the optimization problem (15) we use the Karush–Kuhn–Tucker theorem. We add a dual set of variables, one for each constraint and get the Lagrangian of the optimization problem

$$\begin{aligned}
L(\boldsymbol{W}, \xi, \boldsymbol{\alpha}, \boldsymbol{\lambda}) = & \frac{1}{2}\sum_{m=1}^{k} \boldsymbol{w}_m^T \boldsymbol{w}_m + C\sum_{i=1}^{l} \xi_i - \sum_{i=1}^{l} \lambda_i \xi_i \\
& - \sum_{i=1}^{l} \alpha_i \left(\boldsymbol{w}_{y_i}^T \boldsymbol{\varphi}(\boldsymbol{x}_i) - \frac{1}{k-1}\sum_{m\neq y_i} \boldsymbol{w}_m^T \boldsymbol{\varphi}(\boldsymbol{x}_i) - 1 + \xi_i \right).
\end{aligned} \tag{16}$$

By differentiating the Lagrangian with respect to the primal variables, we obtain the following constraints that the variables have to fulfill in order to be an optimal solution:

$$\frac{\partial L}{\partial \boldsymbol{w}_m} = 0 \iff \boldsymbol{w}_m = \sum_{i: y_i = m} \alpha_i \boldsymbol{\varphi}(\boldsymbol{x}_i) - \frac{1}{k-1}\sum_{i: y_i \neq m} \alpha_i \boldsymbol{\varphi}(\boldsymbol{x}_i), \tag{17}$$

$$\frac{\partial L}{\partial \boldsymbol{\xi}} = 0 \iff C\boldsymbol{e} = \boldsymbol{\lambda} + \boldsymbol{\alpha}, \tag{18}$$

subject to the constraints $\boldsymbol{\alpha} \geqslant \boldsymbol{0}$ and $\boldsymbol{\lambda} \geqslant \boldsymbol{0}$.

By elimination of $\boldsymbol{w}_m$, ξ and $\boldsymbol{\lambda}$, the dual problem of (15) is reduced to the following succinct form

$$\begin{aligned}
\min_{\boldsymbol{\alpha} \in \mathcal{R}^l} \quad & \frac{1}{2}\boldsymbol{\alpha}^T \boldsymbol{G} \boldsymbol{\alpha} - \boldsymbol{e}^T \boldsymbol{\alpha}, \\
\text{subject to} \quad & \boldsymbol{0} \leqslant \boldsymbol{\alpha} \leqslant C\boldsymbol{e}.
\end{aligned} \tag{19}$$

The Hession $\boldsymbol{G}$ is an $l \times l$ matrix with its entries

$$G_{i,j} = \begin{cases} \frac{k}{k-1} K_{i,j}, & \text{if } y_i = y_j, \\ \frac{-k}{(k-1)^2} K_{i,j}, & \text{if } y_i \neq y_j, \end{cases} \tag{20}$$

where we abbreviate the kernel value $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) \equiv \boldsymbol{\varphi}(\boldsymbol{x}_i)^T \boldsymbol{\varphi}(\boldsymbol{x}_j)$ for $K_{i,j}$. Furthermore, let $\boldsymbol{V}$ be an $l \times k$ matrix defined as:

Table 1
The rate of f_{y_i} be the maximum value of $f_1, \ldots, f_k$ for the 10 datasets.

Dataset	Iris	Wine	Glass	Vowel	Segment
Rate	98.67%	99.44%	89.25%	100.00%	99.70%
Dataset	Waveform	DNA	Satimage	Letter	USPS
Rate	94.20%	98.45%	99.66%	99.97%	99.93%

$$V_{i,j} = \begin{cases} 1, & \text{if } y_i = j, \\ \frac{-1}{k-1}, & \text{if } y_i \neq j. \end{cases} \tag{21}$$

We have

$$\boldsymbol{G} = \boldsymbol{K} \odot (\boldsymbol{V}\boldsymbol{V}^T). \tag{22}$$

Since the kernel matrix $\boldsymbol{K}$ and $\boldsymbol{V}\boldsymbol{V}^T$ are both positive semi-definite, so is their Hadamard product $\boldsymbol{G}$. In the next section, we will show that, up to a change of variables, (19) is equivalent to the dual problem of Crammer and Singer's approach with an additional constraint.

In order to avoid the division operation in the kernel computation of (20), we further let $\bar{\boldsymbol{\alpha}} = \frac{k}{(k-1)^2}\boldsymbol{\alpha}$. Therefore, we rewrite (19) as

$$\begin{aligned} \min_{\bar{\boldsymbol{\alpha}} \in \mathcal{R}^l} \quad & \frac{1}{2}\bar{\boldsymbol{\alpha}}^T \overline{\boldsymbol{G}} \bar{\boldsymbol{\alpha}} - \boldsymbol{e}^T \bar{\boldsymbol{\alpha}}, \\ \text{subject to} \quad & 0 \leqslant \bar{\boldsymbol{\alpha}} \leqslant \overline{C}, \end{aligned} \tag{23}$$

where $\overline{C} = \frac{k}{(k-1)^2}C$, and

$$\overline{G}_{i,j} = \begin{cases} (k-1)K_{i,j}, & \text{if } y_i = y_j, \\ -K_{i,j}, & \text{if } y_i \neq y_j. \end{cases} \tag{24}$$

As for the decision function, from (17) we obtain

$$\begin{aligned} f_m^*(\boldsymbol{x}) &= \sum_{i:y_i=m} \alpha_i^* \kappa(\boldsymbol{x}_i, \boldsymbol{x}) - \frac{1}{k-1} \sum_{i:y_i \neq m} \alpha_i^* \kappa(\boldsymbol{x}_i, \boldsymbol{x}) \\ &= \frac{k}{k-1} \sum_{i:y_i=m} \alpha_i^* \kappa(\boldsymbol{x}_i, \boldsymbol{x}) - \underbrace{\frac{1}{k-1} \sum_{i=1}^{l} \alpha_i^* \kappa(\boldsymbol{x}_i, \boldsymbol{x})}_{\text{the same for all } f_m^*}. \end{aligned} \tag{25}$$

Finally, the resulting decision function can be simplified as

$$\arg\max_m f_m^*(\boldsymbol{x}) = \arg\max_m \sum_{i:y_i=m} \alpha_i^* \kappa(\boldsymbol{x}_i, \boldsymbol{x}). \tag{26}$$

Thus, each class model is built upon its own support vectors.

4. Discussions

We discuss the key properties of our proposed SimMSVM approach in this section.

4.1. Relation to the binary SVM

In binary case where $k = 2$, from (17), the SimMSVM approach will produce $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ where $\boldsymbol{w}_1 = -\boldsymbol{w}_2$. Let $\boldsymbol{w} = 2\boldsymbol{w}_1$ and target +1 for the positive class and −1 negative, (15) is equivalent to

$$\begin{aligned} \min_{\boldsymbol{w} \in \mathcal{H}, \xi \in \mathcal{R}^l} \quad & \frac{1}{2}\boldsymbol{w}^T\boldsymbol{w} + 2C\sum_{i=1}^{l} \xi_i, \\ \text{subject to} \quad & y_i \boldsymbol{w}^T \boldsymbol{x}_i \geqslant 1 - \xi_i, \\ & \xi_i \geqslant 0, \quad i = 1, \ldots, l, \end{aligned} \tag{27}$$

which is the unbiased binary SVM formulation.

Lee et al. (2001) proposed a variant of multi-class SVM by modifying the target values so that the positive class has target +1 and the negative class has target $-\frac{1}{k-1}$. For notational convenience, we define $\boldsymbol{v}_m$ for $m = 1, \ldots, k$ as a k-dimensional vector with 1 in the mth coordinate and $-\frac{1}{k-1}$ elsewhere. If the label of a training example $\mathbf{x}_i$ is coded in vector form as $\boldsymbol{v}_{y_i}$, the margin constraints of (15) can be rewritten as

$$\boldsymbol{v}_{y_i}^T \boldsymbol{W}^T \boldsymbol{\varphi}(\boldsymbol{x}_i) \geqslant 1 - \xi_i, \tag{28}$$

where $\boldsymbol{W} = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_k]$. Thus, it can be found that SimMSVM can be degenerated into the unbiased binary SVM.

Note that the transpose of the label vector $\boldsymbol{v}_{y_i}$ is exactly the ith row of $\boldsymbol{V}$ defined in (21). From (22), the Hessian $\boldsymbol{G}$ contains not only a similarity measure between data samples but also their label information. In addition, we find that SimMSVM can be easily extended to the multi-label classification framework in which an instance may be simultaneously relevant to several labels. Multi-label classification is useful for text categorization, multimedia retrieval, and many other areas. In SimMSVM, we solve a multi-label classification problem by using the same dual formulation (19) except for a different representation of the label vector $\boldsymbol{v}_{y_i}$. For example, if a training example is relevant to r out of k labels simultaneously, we can set its label vector so that the relevant labels have target $+\frac{1}{r}$ and the other labels $-\frac{1}{k-r}$, where the values of the k labels are taken into two categories with $+\frac{1}{r}$ and $-\frac{1}{k-r}$. We then compute the Hessian $\boldsymbol{G}$ from (21) and solve the dual problem (19) to obtain the optimal $\boldsymbol{\alpha}^*$.

The resulting discriminant function is given as follows

$$f_m^*(\boldsymbol{x}) = \sum_{i=1}^{l} v_{y_i,m} \alpha_i^* \kappa(\boldsymbol{x}_i, \boldsymbol{x}), \tag{29}$$

where $v_{y_i,m}$ is the mth element of $\boldsymbol{v}_{y_i}$. Since the values of the k labels are taken into two categories with $+\frac{1}{r} > 0$ and $-\frac{1}{k-r} < 0$, the label set Y for the test sample $\boldsymbol{x}$ is determined as:

$$Y = \{m | f_m^*(\boldsymbol{x}) > 0, \ m \in \{1 \ldots k\}\}. \tag{30}$$

4.2. Relation to Crammer and Singer's multi-class SVM

In (Crammer and Singer, 2001), the dual problem is given as follow

$$\begin{aligned} \min_{\boldsymbol{\alpha} \in \mathcal{R}^{l \times k}} \quad & \frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l}\sum_{m=1}^{k} K_{i,j}\alpha_{i,m}\alpha_{j,m} - \sum_{i=1}^{l} \alpha_{i,y_i}, \\ \text{subject to} \quad & \sum_{m=1}^{k} \alpha_{i,m} = 0, \\ & \alpha_{i,m} \leqslant 0 \quad \text{for all} \quad m \neq y_i, \\ & \alpha_{i,y_i} \leqslant C, \\ & i = 1, 2, \ldots, l, \quad m = 1, \ldots, k. \end{aligned} \tag{31}$$

Let $\boldsymbol{M}$ be the $l \times k$ dual matrix whose (i,m)th element is the dual variable $\alpha_{i,m}$, and let

$$\boldsymbol{\alpha} = [\alpha_{1,y_1}, \alpha_{2,y_2}, \ldots, \alpha_{l,y_l}]^T. \tag{32}$$

We then rewrite the dual objective function of (31) as

$$\frac{1}{2}\mathrm{tr}(\boldsymbol{M}^T\boldsymbol{K}\boldsymbol{M}) - \boldsymbol{e}^T\boldsymbol{\alpha}. \tag{33}$$

From the linear constrain of (31), every k dual variables associated with the same x_i follow the sum-to-zero constraint. If $\alpha_{i,y_i} = 0$ for some $\boldsymbol{x}_i$, we have $\alpha_{i,m} = 0$ for all $m \neq y_i$. We carry over the notion of support vectors to the multi-class setting, and define support vectors as examples with $[\alpha_{i,1}, \ldots, \alpha_{i,k}] \neq \mathbf{0}$. A support vector $\boldsymbol{x}_i$ plays a positive role in its class model since $\alpha_{i,yi} > 0$, while punishing some other classes for those $\alpha_{i,m} < 0$. In order to reduce the size of dual variables, we further add to (31) a seemingly aggressive constraint that all those $\alpha_{i,m}$ $(m \neq y_i)$ corresponding to the ith sample are of the same value (share equal punishment):

$$\alpha_{i,m} = -\frac{1}{k-1}\alpha_{i,y_i}, \quad \text{for all} \quad m \neq y_i. \tag{34}$$

Thus, the sum-to-zero constraint of (31) is satisfied accordingly. As for the objective function of (31), from (34) we have

$$\sum_{m=1}^{k}\alpha_{i,m}\alpha_{j,m}=\begin{cases}\frac{k}{k-1}\alpha_i\alpha_j, & \text{if } y_i=y_j,\\ \frac{-k}{(k-1)^2}\alpha_i\alpha_j, & \text{if } y_i\neq y_j,\end{cases} \tag{35}$$

where we simply denote α_{i,y_i} as α_i, for $i=1,\ldots,l$.

To sum up, Crammer and Singer's dual formulation (31) with additional constraint (34) is exactly the same as (19), which is the dual of SimMSVM.

4.3. Fisher consistency issue

In binary case where the class label $y\in\{-1,+1\}$, we denote $P_+(\boldsymbol{x})=P(Y=1|\boldsymbol{X}=\boldsymbol{x})$. Then a binary loss function $\xi(f(\boldsymbol{x}),y)$ is Fisher consistent if the minimizer of $E[\xi(f(\boldsymbol{X}),(Y)|\boldsymbol{X}=\boldsymbol{x})]$ is the same as $\operatorname{sign}(P_+(\boldsymbol{x})-\frac{1}{2})$. In other words, Fisher consistency requires the loss function asymptotically yields the Bayes decision boundary (Lee et al., 2004; Bartlett et al., 2006; Liu, 2007). In the multi-class setting where $y\in\{1,\ldots,k\}$, we let $P_m(\boldsymbol{x})=P(Y=j|\boldsymbol{X}=\boldsymbol{x})$. Suppose $\xi(\boldsymbol{f}(\boldsymbol{x}),y)$ is a multi-class loss function, we denote the minimizer of $E(\xi(\boldsymbol{f}(\boldsymbol{X},Y)|\boldsymbol{X}=\boldsymbol{x}))$ as $\boldsymbol{f}^*=(f_1^*,\ldots,f_k^*)$. Fisher consistency requires $\operatorname{argmax}_m f_m^*=\operatorname{argmax}_m P_m$.

In the next lemma, we will show that, the proposed loss (14) is Fisher inconsistent. For remedy, we will introduce additional constraints to force our loss function to be Fisher consistent.

Lemma 1. *The proposed loss (14) is Fisher inconsistent when $k\geqslant 3$.*

Proof. From the above Fisher consistency definition, we have

$$E\left\{\left[1-f_Y(\boldsymbol{X})+\frac{1}{k-1}\sum_{M\neq Y}f_M(\boldsymbol{X})\right]_+\right\}$$
$$=E\left\{\sum_{y=1}^{k}P_y(\boldsymbol{X})\left[1-f_y(\boldsymbol{X})+\frac{1}{k-1}\sum_{m\neq y}f_m(\boldsymbol{X})\right]_+\right\}.$$

For any fixed $X=x$, the goal is to minimize

$$\sum_{y=1}^{k}P_y(\boldsymbol{x})\left[1-f_y(\boldsymbol{x})+\frac{1}{k-1}\sum_{m\neq y}f_m(\boldsymbol{x})\right]_+. \tag{36}$$

We let

$$g_y(\boldsymbol{x})=f_y(\boldsymbol{x})-\frac{1}{k-1}\sum_{m\neq y}f_m(\boldsymbol{x}), \tag{37}$$

for $y=1,\ldots,k$. Therefore, (36) is equivalent to

$$\sum_{y=1}^{k}P_y(\boldsymbol{x})\left[1-g_y(\boldsymbol{x})\right]_+. \tag{38}$$

Clearly, $\sum_{y=1}^{k}g_y(\boldsymbol{x})=0$. From the proof of Lemma 1 in (Liu, 2007), we know that the minimizer g^* satisfies $g_m^*\leqslant 1,\forall m=1,\ldots,k$. Thus, the problem (36) reduces to

$$\begin{aligned}\max_{\boldsymbol{g}}\quad & \sum_{m=1}^{k}P_m(\boldsymbol{x})g_m(\boldsymbol{x}),\\ \text{subject to}\quad & \sum_{m=1}^{k}g_m(\boldsymbol{x})=0,\\ & g_m(\boldsymbol{x})\leqslant 1,\quad m=1,\ldots,k.\end{aligned} \tag{39}$$

It is easy to verify that the solution of (39) satisfies $g_m^*(\boldsymbol{x})=-(k-1)$ if $m=\arg\min_j P_j(\mathbf{x})$ and 1 otherwise. We then rewrite (37) as

$$f_y(\boldsymbol{x})=\frac{k-1}{k}(g_y(\boldsymbol{x})+A), \tag{40}$$

where $A=\frac{1}{k-1}\sum_{m=1}^{k}f_m(\boldsymbol{x})$. We have

$$\operatorname{argmax}_m f^*(\boldsymbol{x})=\operatorname{argmax}_m g^*(\boldsymbol{x}). \tag{41}$$

Therefore, our loss function (14) is not Fisher consistent when $k\geqslant 3$. □

Although Fisher consistency is a desirable condition, a consistent loss may not always lead to better classification accuracy (Hsu and Lin, 2002b; Rifkin and Klautau, 2004; Liu, 2007). However, if Fisher consistency is required for our loss function, one alternative solution is to add further constraints

$$(k-1)f_y(\boldsymbol{x})-\sum_{m\neq y}f_m(\boldsymbol{x})+1\geqslant 0, \tag{42}$$

for $y=1,\ldots,k$. These constraints (42) amount to $g_m(\boldsymbol{x})\geqslant -\frac{1}{k-1}$. We have the following Lemma:

Lemma 2. *The maximizer g^* of $E[g_Y(X)|X=x]$, subject to $\sum_{m=1}^{k}g_m(\boldsymbol{x})=0$ and $-\frac{1}{k-1}\leqslant g_m(\boldsymbol{x})\leqslant 1,\forall m$, satisfies the following: $g_m^*(\boldsymbol{x})=1$ if $m=\operatorname{argmax}_m P_m(x)$ and $-\frac{1}{k-1}$ otherwise.*

For the detailed proof, please see Lemma 5 in (Liu, 2007). □

Lemma 3. *Justifies that, with additional constraints (42), the proposed loss function will be Fisher consistent. Fisher consistency is an attractive property for a loss function, we will study the Fisher consistency issue in more detail in our future work.*

5. Experiments

In this section, we validate the accuracy and efficiency of the proposed SimMSVM algorithm on several multi-class pattern recognition problems including **Iris**, **Wine**, **Glass**, **Vowel**, **Segment**, and **Waveform** from the **UCI** repository (Asuncion and Newman, 2007); **DNA**, **Satimage**, **Letter** from **Statlog** collection (King et al., 1995); and **USPS** (Hull, 1994). For all the datasets except **DNA**, we linearly scale each attribute to be in the range $[-1,1]$. Table 2 gives a detailed description about these datasets. Here we compare the proposed SimMSVM with some classical multi-class algorithms in terms of classification and computation time. Moreover, we also analyze the scaling behavior of the proposed SimMSVM with the sample size and the number of classes in terms of training time. All the experiments are performed using the tool of BSVM (Hsu and Lin, 2008) on a 1.50 GHz Intel Core^{TM}2 Duo PC running Windows XP with 1 GB main memory.

5.1. Classification performance comparisons

The tool of BSVM is used in our experiments. BSVM is developed to efficiently solve the bound-constrained problem (4). We find that the only difference between (19) and (4) is the Hessian of the objective function. To proceed the proposed SimMSVM algorithm and take advantage of its sophisticated implementations,

Table 2
Dataset description.

Dataset	# Training Data (l)	# Test Data	# Class (k)	# Attr (d)
Iris	150	0	3	4
Wine	178	0	3	13
Glass	214	0	6	9
Vowel	528	0	11	10
Segment	2310	0	7	19
Waveform	5000	0	3	21
DNA	2000	1186	3	180
Satimage	4435	2000	6	36
Letter	15,000	5000	26	16
USPS	7291	2007	10	256

Table 3
Classification accuracy (%) and the number of SV comparison between 1VR, 1V1, W&W, C&S and SimMSVM (best rates bold-faced).

Dataset	1VR		1V1		W&W		C&S		SimMSVM	
	ACC	SV	ACC	SV	ACC	SV	ACC	SV	ACC	SV
Iris	95.47 ±5.62	43.0 ±17.5	95.47 ±5.46	40.1 ±24.5	96.00 ±5.04	33.3 ±18.8	95.33 ±5.08	35.7 ±23.6	**96.53** ±4.52	18.0 ±15.4
Wine	96.82 ±4.15	64.2 ±25.9	**97.76** ±3.74	70.8 ±17.4	97.53 ±3.58	64.9 ±28.4	97.06 ±3.41	64.0 ±36.4	97.18 ±3.80	101.6 ±15.0
Glass	65.90 ±9.69	140.7 ±10.7	68.67 ±10.50	128.8 ±8.2	67.81 ±10.02	141.3 ±19.2	69.05 ±10.77	152.8 ±17.1	**71.62** ±10.36	136.6 ±4.2
Vowel	97.88 ±2.37	348.2 ±5.5	**99.31** ±1.27	346.0 ±4.8	98.65 ±1.61	312.8 ±60.8	98.88 ±1.61	367.1 ±64.0	99.08 ±1.24	403.9 ±4.1
Segment	96.73 ±0.93	483.5 ±43.9	97.23 ±0.96	444.4 ±53.7	97.07 ±0.87	338.5 ±57.8	97.31 ±0.83	626.1 ±384.8	**97.33** ±0.91	1106.3 ±16.5
Waveform	86.38 ±1.77	1843.6 ±180.7	**87.06** ±1.16	2374.6 ±302.1	87.00 ±1.17	2187.1 ±285.3	86.75 ±1.52	2061.2 ±149.9	86.64 ±1.32	2870.0 ±8.5
DNA	93.79 ±0.59	808.4 ±10.3	94.69 ±0.32	734.4 ±84.4	94.77 ±0.34	708.2 ±72.6	**94.95** ±0.31	660.2 ±37.3	93.81 ±0.23	458.6 ±4.7
Satimage	89.20 ±0.44	1312.2 ±18.9	**91.21** ±0.50	1233.2 ±23.8	90.72 ±0.31	1122.8 ±34.8	90.60 ±0.24	1467.4 ±735.3	90.57 ±0.41	2731.0 ±8.6
Letter	94.01 ±0.24	4464.4 ±13.9	**97.04** ±0.16	5439.4 ±22.7	96.52 ±0.21	3541.8 ±30.6	96.39 ±0.17	3424.8 ±21.1	96.37 ±0.14	8126.2 ±27.3
USPS	93.49 ±0.27	1348.6 ±14.4	94.96 ±0.22	1247.2 ±10.8	94.83 ±0.41	974.4 ±20.2	94.80 ±0.52	2486.2 ±138.2	**95.00** ±0.37	3349.2 ±17.5

Table 4
Testing time (in second) and the number of SV comparison between W&W, C&S and SimMSVM.

Dataset (# class)	W&W			C&S			SimMSVM		
	Time	SV	$\frac{\text{Time}}{\text{SV}}$	Time	SV	$\frac{\text{Time}}{\text{SV}}$	Time	SV	$\frac{\text{Time}}{\text{SV}}$
Iris (3)	2.44×10^{-4} $\pm 1.34 \times 10^{-4}$	33.3 ±18.8	7.31×10^{-6}	3.69×10^{-4} $\pm 2.44 \times 10^{-4}$	35.7 ±23.6	10.34×10^{-6}	0.63×10^{-4} $\pm 0.54 \times 10^{-4}$	18.0 ±15.4	3.52×10^{-6}
Wine (3)	5.31×10^{-4} $\pm 2.30 \times 10^{-4}$	64.9 ±28.4	8.19×10^{-6}	7.48×10^{-4} $\pm 4.26 \times 10^{-4}$	64.0 ±36.4	11.69×10^{-6}	4.08×10^{-4} $\pm 0.64 \times 10^{-4}$	101.6 ±15.0	4.01×10^{-6}
Glass (6)	35.40×10^{-4} $\pm 4.89 \times 10^{-4}$	141.3 ±19.2	25.07×10^{-6}	44.30×10^{-4} $\pm 4.91 \times 10^{-4}$	152.8 ±17.1	28.97×10^{-6}	6.75×10^{-4} $\pm 0.21 \times 10^{-4}$	136.6 ±4.2	4.94×10^{-6}
Vowel (11)	3.87×10^{-2} $\pm 0.76 \times 10^{-2}$	312.8 ±60.8	12.37×10^{-5}	4.83×10^{-2} $\pm 0.84 \times 10^{-2}$	367.1 ±64.0	13.17×10^{-5}	0.49×10^{-2} $\pm 0.00 \times 10^{-2}$	403.9 ±4.1	1.21×10^{-5}
Segment (7)	1.12×10^{-1} $\pm 0.19 \times 10^{-1}$	338.5 ±57.8	32.98×10^{-5}	2.34×10^{-1} $\pm 1.44 \times 10^{-1}$	626.1 ±384.8	37.32×10^{-5}	0.60×10^{-1} $\pm 0.01 \times 10^{-1}$	1106.3 ±16.5	5.39×10^{-5}
Waveform (3)	0.52 ±0.07	2187.1 ±285.3	2.4×10^{-4}	0.72 ±0.05	2061.2 ±149.9	3.4×10^{-4}	0.34 ±0.00	2870.0 ±8.5	1.2×10^{-4}
DNA (3)	0.63 ±0.01	708.2 ±72.6	9×10^{-4}	0.82 ±0.01	660.2 ±37.3	12×10^{-4}	0.28 ±0.00	458.6 ±4.7	6×10^{-4}
Satimage (6)	2.66 ±0.31	1122.8 ±34.8	24×10^{-4}	3.07 ±0.24	1467.4 ±735.3	21×10^{-4}	1.28 ±0.41	2731.0 ±8.6	5×10^{-4}
Letter (26)	107.1 ±0.6	3541.8 ±30.6	30×10^{-3}	103.7 ±0.8	3424.8 ±21.1	30×10^{-3}	9.5 ±0.0	8126.2 ±27.3	1×10^{-3}
USPS (10)	4.19 ±0.07	974.4 ±20.2	43×10^{-4}	16.46 ±0.17	2486.2 ±138.2	66×10^{-4}	1.59 ±0.01	*3349.2* ±17.5	4×10^{-4}

we only need to modify the kernel evaluation part of BSVM. For the one-versus-one (1V1) and one-versus-rest (1VR) approaches, we formulate their binary subproblems by (3) and solve each of them using BSVM. For the two so-called "all-together" methods proposed by Weston and Watkins and Crammer and Singer, their major disadvantage remains to the enormous size of the optimization problems. To tackle these difficulties, Hsu and Lin proposed decomposition methods for these two approaches, respectively. The implementations are integrated into BSVM as well.

In conclusion, we adopt BSVM for three reasons: (1) BSVM uses a simple working set selection which leads to a faster convergence for difficult cases. The use of a special implementation of the optimization solver allows BSVM to stably identify bounded variables. (2) The implementations of Weston & Watkins' approach (Weston and Watkins (1999)) and Crammer and Singer's approach (Crammer and Singer, 2001) are integrated into BSVM. We can formulate the binary subproblems of the one-versus-one and one-versus-rest approaches and solve each of them using BSVM. (3) For the proposed SimMSVM algorithm, we just modify the kernel evaluation part of BSVM. Furthermore, it should be stated that as described in Hsu and Lin (2002a), the BSVM software includes a decomposition implementation of Crammer and Singer's approach Crammer and Singer (2001) for multi-class problems. The other three approaches (one-versus-one, one-versus-rest, and Weston and Watkins' approach) are also based on BSVM for the fair comparison. Therefore, the experiments of this manuscript are all implemented in a fair condition.

The most important criterion for evaluating the performance of the multi-class SVM is the classification accuracy. For all the five methods, we use the gaussian RBF kernel

$$\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp(-\gamma\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2), \quad \gamma \geqslant 0.$$

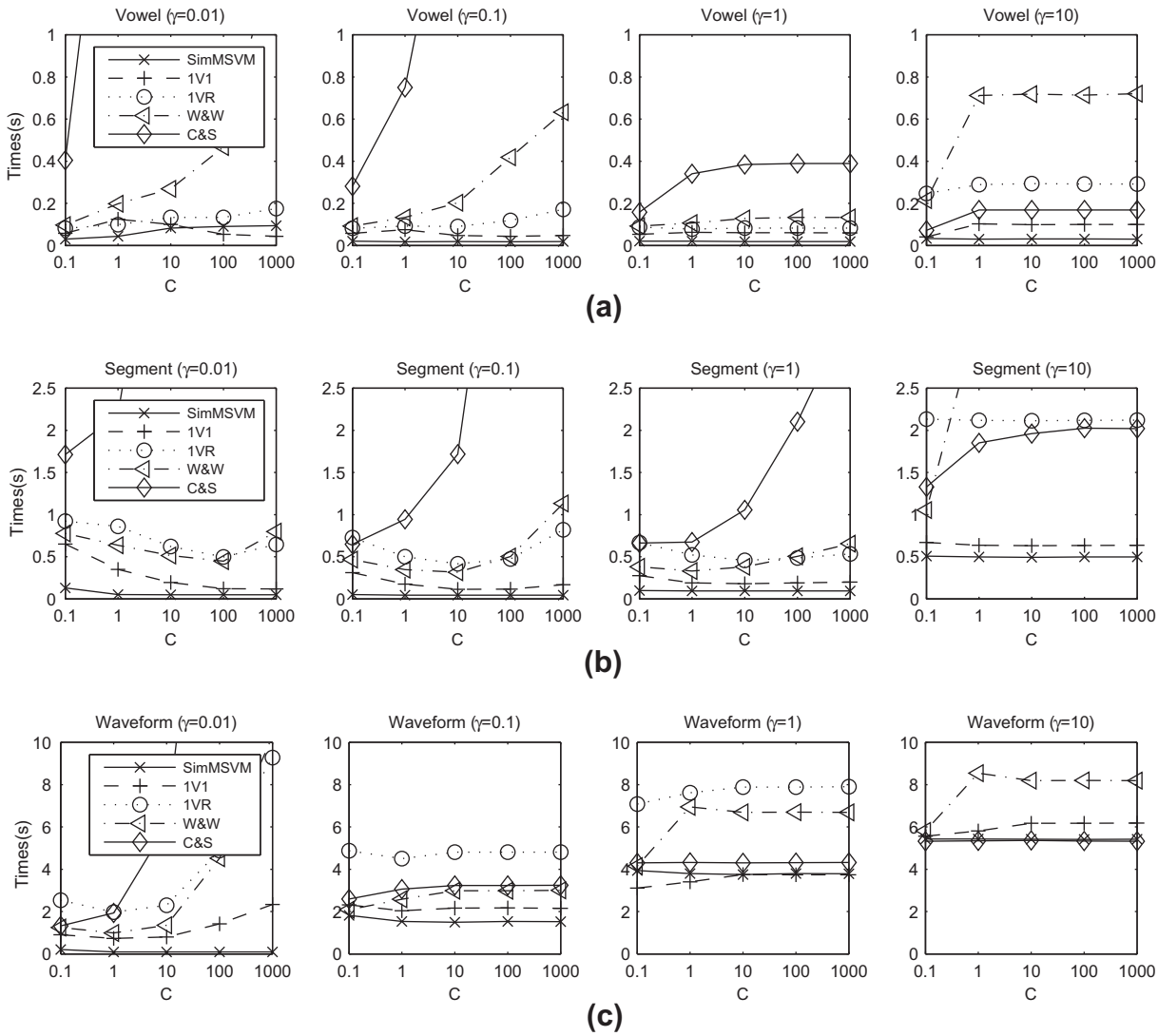

Fig. 2. Cross-validation time (in seconds) comparison as a function of C for $\gamma = 0.01$, 0.1, 1, and 10 on (a) vowel dataset, (b) segment dataset, (c) waveform dataset.

We follow the strategy used in (Hsu and Lin, 2002a), and use model selection to get the optimal parameters. The hyper-parameter space is explored on a two dimensional grid with $\gamma = [10^{-2}, 10^{-1}, 10^{0}, 10^{1}]$ and the regularization parameter $C = [10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}]$. We use two criteria to estimate the generalized accuracy.

For the used **Iris**, **Wine**, **Glass**, **Vowel**, **Segment**, and **Waveform**, we randomly subdivide each dataset into 10 disjoint subsets. Then for each procedure, one of the subsets is used for testing and the others for training, where the 10-fold cross-validation strategy is employed in the training set so as to select the optimal parameter C and γ. In order to evaluate the variability over different random splits for each dataset, the whole procedure above is repeated 5 times and the final experimental results are averaged. For the used **DNA**, **Satimage**, **Letter**, and **USPS** where both training and testing sets are available, we randomly draw the subsets of the original training set for training in order to evaluate the variability over different choices of the training set on each dataset. Here the 70% data points are randomly chosen from the original training set, where the 10-fold cross-validation is used for parameter selection on the selected training set. Then we train on the 70% dataset with the optimal pair (C, γ) and predict for the test set. The whole procedure above is repeated 5 times and the results are averaged. We present the comparison results in Table 3, where "W&W" denotes the method proposed by Weston and Watkins, and "C&S" indicates Crammer and Singers approach.

Table 3 shows the optimal classification accuracy and their corresponding number of support vectors (SV). Both the average standard deviation of the classification accuracy and the number of SV are also reported in Table 3 so as to evaluate the statistical significance of the cross-validation results and the number of SV are not integers since they are the average values of the 10-fold cross-validation. From Table 3, we can find that the proposed simplified method gives competitive accuracies with the method by Crammer and Singer. In detail, we observe that: (1) not all the 1VR and 1V1 approaches outperform "all together" approaches (i.e. W&W, C&S, and SimMSVM) in the used ten datasets; (2) all the five implemented methods give the comparable results. The experimental results in the previous work such as Hsu and Lin (2002a) also suggest that no one is statistically better than the others for the 1VR and 1V1 approaches and all together approaches.

Since the test time of the W&W, C&S, and SimMSVM depends on the number of SV, we analyze their decision function expressions. Suppose that there are kernel function $\kappa(\boldsymbol{x}_i, \boldsymbol{x})$ and let k and n_{SV} be the number of class and SVs. The computational complexity of Eq.

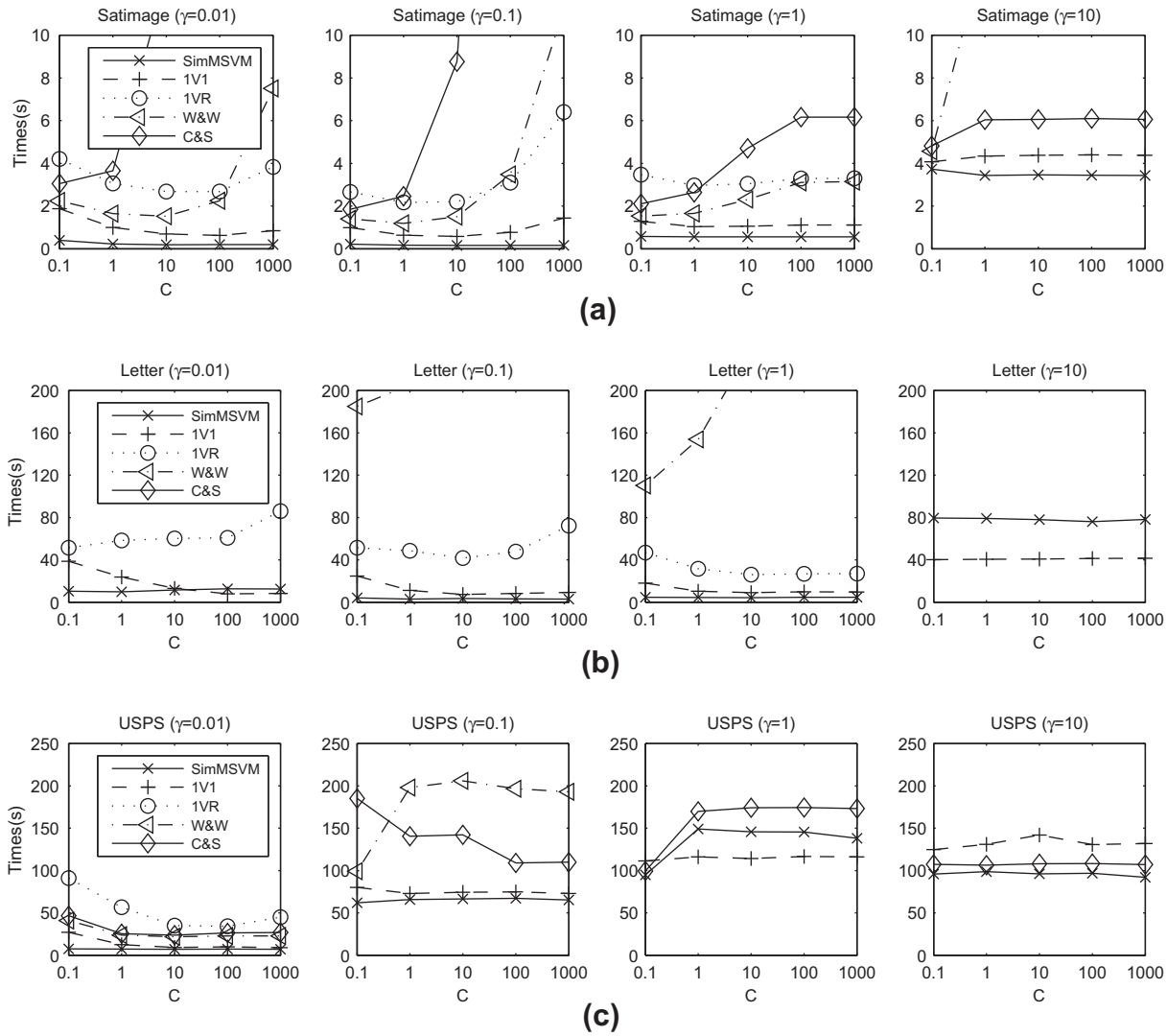

Fig. 3. Cross-validation time (in seconds) comparison as a function of C for $\gamma = 0.01$, 0.1, 1, and 10 on (a) satimage dataset, (b) letter dataset, (c) USPS dataset.

(25) (i.e. $\arg\max_m f_m^*(\boldsymbol{x}) = \arg\max_m \sum_{i:y_i=m} \alpha_i^* \kappa(\boldsymbol{x}_i, \boldsymbol{x})$) is $O(n_{SV})$ since each SV is operated once for testing. According to Hsu and Lin (2002a), Eq. (6) (i.e. $\arg\max_m f_m(\boldsymbol{x}) = \arg\max_m (\boldsymbol{w}_m^T \boldsymbol{\varphi}(\boldsymbol{x}) + b_m)$) can be rewritten as

$$\arg\max_m f_m(\boldsymbol{x}) = \arg\max_m \sum_{i=1}^{l} (c_i^m A_i - \alpha_i^m)(\kappa(\boldsymbol{x}_i, \boldsymbol{x}) + 1),$$

where $A_i = \sum_m \alpha_i^m$, $c_i^m = 1$, if $y_i = m$ and $c_i^m = 0$, otherwise. Furthermore, Eq. (8) (i.e. $\arg\max_m f_m(\boldsymbol{x}) = \arg\max_m \boldsymbol{w}_m^T \boldsymbol{\varphi}(\boldsymbol{x})$) can be rewritten as

$$\arg\max_m f_m(\boldsymbol{x}) = \arg\max_m \sum_{i=1}^{l} \alpha_i^m \kappa(\boldsymbol{x}_i, \boldsymbol{x}).$$

Therefore, the computational complexity of Eqs. (6) and (8) is $O(kn_{SV})$. A quantitative test time comparison of the used datasets among these methods is also given in Table 4. From this table, the proposed method has a comparable test time to the multi-class SVM methods W&W and C&S.

5.2. Computational complexity comparisons

We have claimed that the proposed SimMSVM achieves a significant speed-up compared with the method by Crammer and Singer for its simplified model. To demonstrate it, we report the average training time of the above model selection procedures. Due to the limit of the space here, we select the six out of ten datasets for comparison, e.g. **Vowel**, **Segment**, **Waveform**, **Satimage**, **Letter**, and **USPS**. For the six datasets, we report the average cross-validation time of the five competitive methods as shown in Figs. 2 and 3. Each row of Figs. 2 and 3 has the parameter γ with the value 10^{-2}, 10^{-1}, 10^{0}, and 10^{1}, respectively, and the C varies from 10^{-1} to 10^{3} in a log scale. For the parameter pair (C,γ) for each learning algorithm, we report the average time on ten runs of the cross-validation. Most of the standard deviation value are below 0.05 in our experiments.

Figs. 2 and 3 show that the proposed SimMSVM has the best computational cost for most of the parameter combinations. For example, the 18 out of 20 parameter combinations in **Vowel** dataset by the SimMSVM outperform the other methods. All the parameter combinations in **Satimage** dataset by SimMSVM lead to better performance. Totally, the 81.7% combinations by SimMSVM are the most efficient one for the used six datasets. In fact, it gives a significant speed-up from Crammer and Singer's approach (95.8% combinations by SimMSVM perform better). Although the BSVM has applied decomposition methods for the two "all-together" approaches, their training speed remains slow especially for the rel-

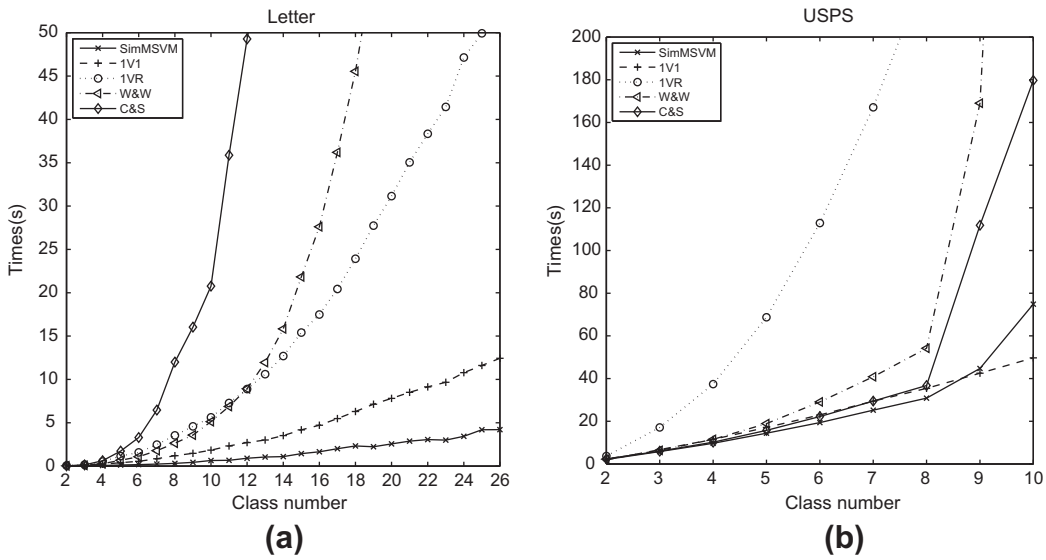

Fig. 4. Training time (seconds) vs. class number on (a) letter dataset, (b) USPS dataset.

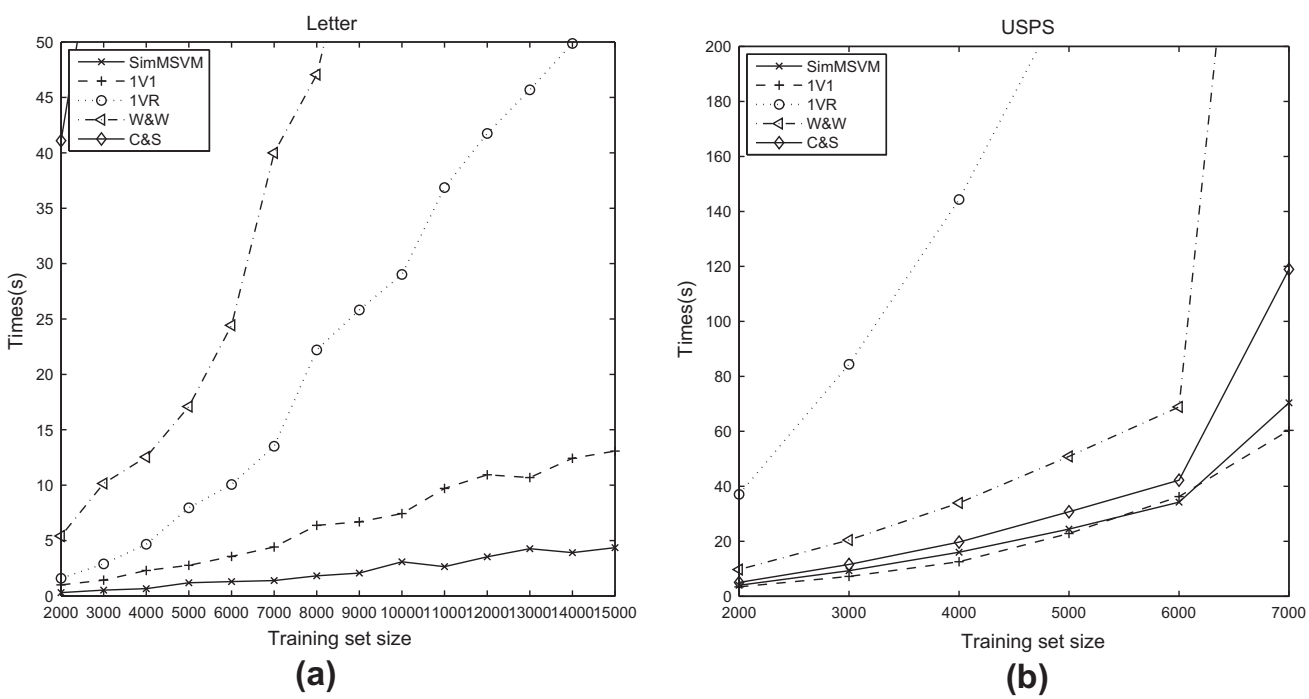

Fig. 5. Training time (seconds) vs. increasing training set size by keeping constant the class number on (a) letter dataset, (b) USPS dataset.

atively small γ. Since we find that the optimal parameters (C,γ) are in various ranges for different classification cases in the experiments, it is necessary to carry out more parameter pairs to obtain the optimal model. If the training speed is an important issue, the proposed SimMSVM could be an option.

5.3. Speed vs. data size and number of class

In this subsection, we give a detailed analysis for the effect of the number of classes and the training set size on the proposed method. We compare the SimMSVM with the other multi-class SVM approaches on two datasets with the large training samples: **Letter** and **USPS**. First, we restrict the analysis to binary-class case for the first two classes, then proceed to a multi-class analysis by gradually adding a new class. As described in Figs. 2 and 3, different kernel parameters (C,γ) would cause different computational time. For fairness, we choose the default parameter values of the BSVM package for the comparison: $C = 1$, $\gamma = 1/k$. Fig. 4 summarizes the results.

For the two "all-together" multi-class SVM approaches, we have to solve an $(l \times k)$-variable QP. For the one-versus-rest approach, instead we need to solve k l-variable QPs. As for the proposed SimMSVM, we solve a single l-variable QP, which is not affected by k. Fig. 4 validates that the training time of SimMSVM scales slowly as the number of classes and data set increase. We show an increasing training set size by keeping the number of class number ($k = 26$ for Letter and $k = 10$ for USPS) constant and adding training samples to each class. Each learning algorithm is repeated with five times and

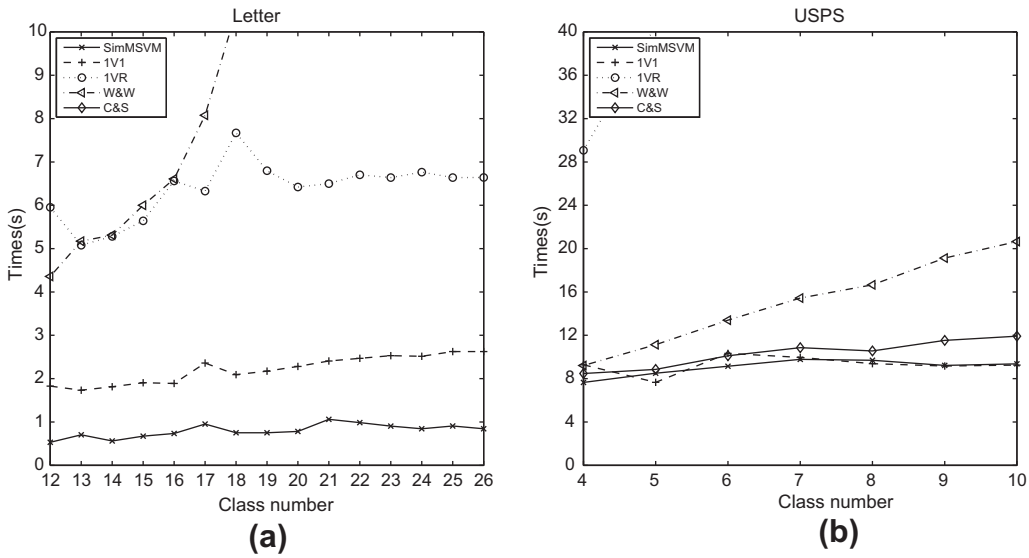

Fig. 6. Training time (seconds) vs. increasing class number by keeping constant the training set size on (a) letter dataset, (b) USPS dataset.

we report the average training time in Fig. 5. We further show an increasing of class number by keeping the training set size constant. In our experiments, we set the constant training set size to be 5000 for Letter and 3000 for USPS. The number of the training samples of each class is reduced accordingly while the class number increases. Fig. 6 summarizes the results.

From Figs. 4–6, it can be found that the proposed SimMSVM takes the least computational cost in most cases except the **USPS**. In the **USPS**, the SimMSVM has a comparable computational cost to that of the one-versus-one method. The reason for this phenomenon is that the optimization for the dual problem of SVM with the 1V1 is much lower than that of the proposed SimMSVM in the large scale case. In this case, the advantage of the SimMSVM might be counteracted with that of the optimization for the traditional SVM dual problem with 1V1. Compared with the other three algorithms (one-versus-rest, the C&S and the W&W), the SimMSVM here has a significant advantage, which can demonstrate the efficiency of SimMSVM.

6. Conclusions

In this paper, we introduce a simplified multi-class SVM algorithm that directly solves a multi-class classification problem. Through modifying Crammer and Singer's multi-class SVM by introducing a relaxed classification error bound, the proposed SimMSVM reduces the size of the dual variables from $l \times k$ to l, where l and k are the size of training data and number of classes, respectively. We also prove that the dual formulation of the proposed SimMSVM is exactly the dual of Crammer and Singer's approach with an additional constraint. The experimental evaluations on several real world datasets show that the proposed SimMSVM approach can greatly speed-up the training process and achieve a competitive classification accuracy.

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